Earners AI and ML by Aditya Jain sir -

Summary Notes

Lecture 13 Recap - Functions

```
def show():
  print("Hello Friends!")
```

```
show()
```

```
Hello Friends!
```

```
show()
```

```
Hello Friends!
```

```
def show(name, age):
  print("name is", name)
  print("age is", age)
```

```
show("Raj", 20)
```

```
name is Raj
age is 20
```

```
show(20, "Raj")
```

```
name is 20
age is Raj
```

```
show(name = "Raj", age = 20)
```

```
name is Raj
age is 20
```

```
show(age = 20, name = "Raj")
```

```
name is Raj
age is 20
```

```
#default argument

def show(name = "Raj"):
  print(name)
```

```
show()
```

```
Raj
```

```
show("AJ")
```

```
AJ
```

HW

```
def check(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"
```

```
print(check(15))
```

```
Odd
```

```
def greet(name):
    print("Hello", name)

greet("Rahul")
```

```
Hello Rahul
```

```
def total(numbers):
    s = 0

    for n in numbers:
        s += n

    return s

print(total([5, 10, 15]))
```

```
30
```

```
def display():
    return "Python"

print(display())
```

```
Python
```

```
def largest(a, b):
    if a > b:
        return a
    else:
        return b

print(largest(12, 25))
```

```
25
```

Dictionary

```
student = ["Rahul",22,91.5,"Nagpur"]
```

```
student = {
    "name" : "Rahul",
    "age" : 22,
    "CGPA" : 91.5,
    "city" : "Nagpur"
}
```

```
student
```

```
{'name': 'Rahul', 'age': 22, 'CGPA': 91.5, 'city': 'Nagpur'}
```

```
type(student)
```

```
dict
```

```
student[1]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_1193/2126975645.py in <cell line: 0>()
----> 1 student[1]

KeyError: 1
```

Next steps: Explain error

```
student["name"]
```

```
'Rahul'
```

```
student["CGPA"]
```

```
91.5
```

```
student["age"] = 23
```

```
print(student)
```

```
{'name': 'Rahul', 'age': 23, 'CGPA': 91.5, 'city': 'Nagpur'}
```

```
student["state"] = "Maharashtra"
```

```
student
```

```
{'name': 'Rahul',
 'age': 23,
 'CGPA': 91.5,
 'city': 'Nagpur',
 'state': 'Maharashtra'}
```

```
student["Age"]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_1193/2431116529.py in <cell line: 0>()
----> 1 student["Age"]

KeyError: 'Age'
```

Next steps: Explain error

```
student["address"]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_1193/1757129455.py in <cell line: 0>()
----> 1 student["address"]

KeyError: 'address'
```

Next steps: Explain error

```
student.get("age")
```

```
23
```

```
student.get("address")
```

```
student.get("address", "Key not found!")
```

```
'Key not found!'
```

```
student.get("age", "Key not found!")
```

```
23
```

```
student
```

```
{'name': 'Rahul',
 'age': 23,
 'CGPA': 91.5,
 'city': 'Nagpur',
 'state': 'Maharashtra'}
```

```
student.pop("state")
```

```
'Maharashtra'
```

```
student
```

```
{'name': 'Rahul', 'age': 23, 'CGPA': 91.5, 'city': 'Nagpur'}
```

```
student.pop("state")
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_1193/2866018373.py in <cell line: 0>()
----> 1 student.pop("state")

KeyError: 'state'
```

Next steps: Explain error

Loop over Dictionaries

```
student
```

```
{'name': 'Rahul', 'age': 23, 'CGPA': 91.5, 'city': 'Nagpur'}
```

```
for k in student.keys():
  print(k)
```

```
name
age
CGPA
city
```

```
for k in student.keys():
  print(student[k])
```

```
Rahul
23
91.5
Nagpur
```

```
for k in student.values():
  print(k)
```

```
Rahul
23
91.5
Nagpur
```

```
for elem in student.items():
  print(elem)
```

```
('name', 'Rahul')
('age', 23)
('CGPA', 91.5)
('city', 'Nagpur')
```

```
for elem in student.items():
  print("key =",elem[0], ", value =", elem[1])
```

```
key = name , value = Rahul
key = age , value = 23
key = CGPA , value = 91.5
key = city , value = Nagpur
```

Quiz

```
student = {
    "name": "Rahul",
```

```
    "age": 22
}

print(student["name"])
```

```
Rahul
```

```
student = {
    "name": "Rahul",
    "age": 22
}

print(student["age"])
```

```
22
```

```
Which of the following correctly creates a dictionary?

A.

student = ["name":"Rahul"]

B.

student = ("name":"Rahul")

C.

student = {"name":"Rahul"}

D.

student = <"name":"Rahul">


Ans: C
```

```
student = {
    "name": "Rahul"
}

student["age"] = 22

print(student)
```

```
{'name': 'Rahul', 'age': 22}
```

```
student = {
    "name": "Rahul",
    "age": 22
}

student["age"] = 23

print(student)
```

```
{'name': 'Rahul', 'age': 23}
```

```
student = {
    "name":"Rahul",
    "age":22
}

print(len(student))
```

```
2
```

```
student = {
    "name":"Rahul"
}

print(student.get("city"))
```

```
None
```

```
student.get("city")
```

```
student = {
    "name":"Rahul"
}

print(student["city"])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_1193/2849176709.py in <cell line: 0>()
      3 }
      4
----> 5 print(student["city"])

KeyError: 'city'
```

Next steps: Explain error

```
student = {
    "name":"Rahul",
    "age":22
}

print(student.keys())
```

```
dict_keys(['name', 'age'])
```

```
student = {
    "name":"Rahul",
    "age":22
}

print(student.values())
```

```
dict_values(['Rahul', 22])
```

```
student = {
    "name":"Rahul",
    "age":22
}

for key in student.keys():
    print(student[key])
```

```
Rahul
22
```

```
student = {
    "name":"Rahul",
    "city":"Nagpur"
}

student.pop("city")

print(student)
```

```
{'name': 'Rahul'}
```

```
marks = {
    "Math":90,
    "Science":80,
    "English":95
}

total = 0

for subject in marks.keys():
    total += marks[subject]

print(total)
```

```
265
```

```
cart = {
    "Milk":2,
    "Bread":1
}

cart["Eggs"] = 12

print(len(cart))
```

```
3
```

```
employee = {
    "salary":50000
}

employee["salary"] += 10000

print(employee["salary"])
```

```
60000
```

```
student = {
    "name":"Rahul",
    "age":22
}

count = 0

for key in student.keys():
    count += 1

print(count)
```

```
2
```

```
student = {
    "name": "Rahul",
    "age": 22
}

student["city"] = "Nagpur"
student["age"] += 1

print(student["age"])
```

```
23
```

```
marks = {
    "Math": 90,
    "Science": 80,
    "English": 70
}

total = 0

for subject in marks.keys():
    total += marks[subject]

print(total / len(marks))
```

```
80.0
```

```
employee = {
    "salary": 50000
}

employee["salary"] = employee["salary"] * 1.10

print(employee["salary"])
```

```
55000.00000000001
```

```
# 10% = 10/100 = 0.1

# x = x + 0.1x = (1 + 0.1)x = 1.1x
```

```
data = {
    "A": 10,
    "B": 20,
    "C": 30
}

count = 0

for key in data.keys():
    if data[key] > 15:
        count += 1

print(count)
```

```
2
```

```
student = {
    "name": "Rahul",
    "age": 22,
    "city": "Nagpur"
}

for key in student.keys():
    print(len(key))
```

```
4
3
4
```

```
for k in student:
  print(k)
```

```
name
age
city
```

HW

```
fruit = {
    "Apple": 50,
    "Banana": 20
}

fruit["Apple"] += fruit["Banana"]

print(fruit["Apple"])
```

```
student = {
    "Math": 90,
    "Science": 85,
    "English": 95
}

highest = 0

for subject in student:
    if student[subject] > highest:
        highest = student[subject]

print(highest)
```

```
student = {
    "name": "Rahul",
    "age": 22
}

student["age"] = student["age"] + len(student)
```

```
print(student["age"])
```

```
data = {}

data["A"] = 10
data["B"] = 20
data["A"] = 30

print(len(data))
```

```
student = {
    "Math": 80,
    "Science": 90,
    "English": 85
}

count = 0

for key in student:
    if student[key] >= 85:
        count += 1

print(count)
```

```
person = {
    "name": "Amit",
    "age": 25
}

person.pop("age")

person["age"] = 30

print(person)
```

```
marks = {
    "Math": 90,
    "Science": 80,
    "English": 70
}

total = 0

for subject in marks:
    if marks[subject] >= 80:
        total += marks[subject]

print(total)
```

```
student = {
    "Math": 95,
    "Science": 85
}

for key in student:
    print(key, student[key] // 10)
```

```
data = {
    "A": 5,
    "B": 10,
    "C": 15
}

result = 0

for key in data:
    if data[key] % 10 == 0:
        result += 1

print(result)
```

```
student = {
    "Math": 90,
    "Science": 80
}

student["Total"] = 0

for key in student:
    if key != "Total":
        student["Total"] += student[key]

print(student["Total"])
```